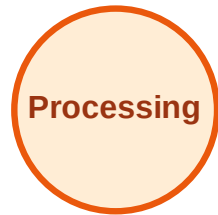


BFS Traversal

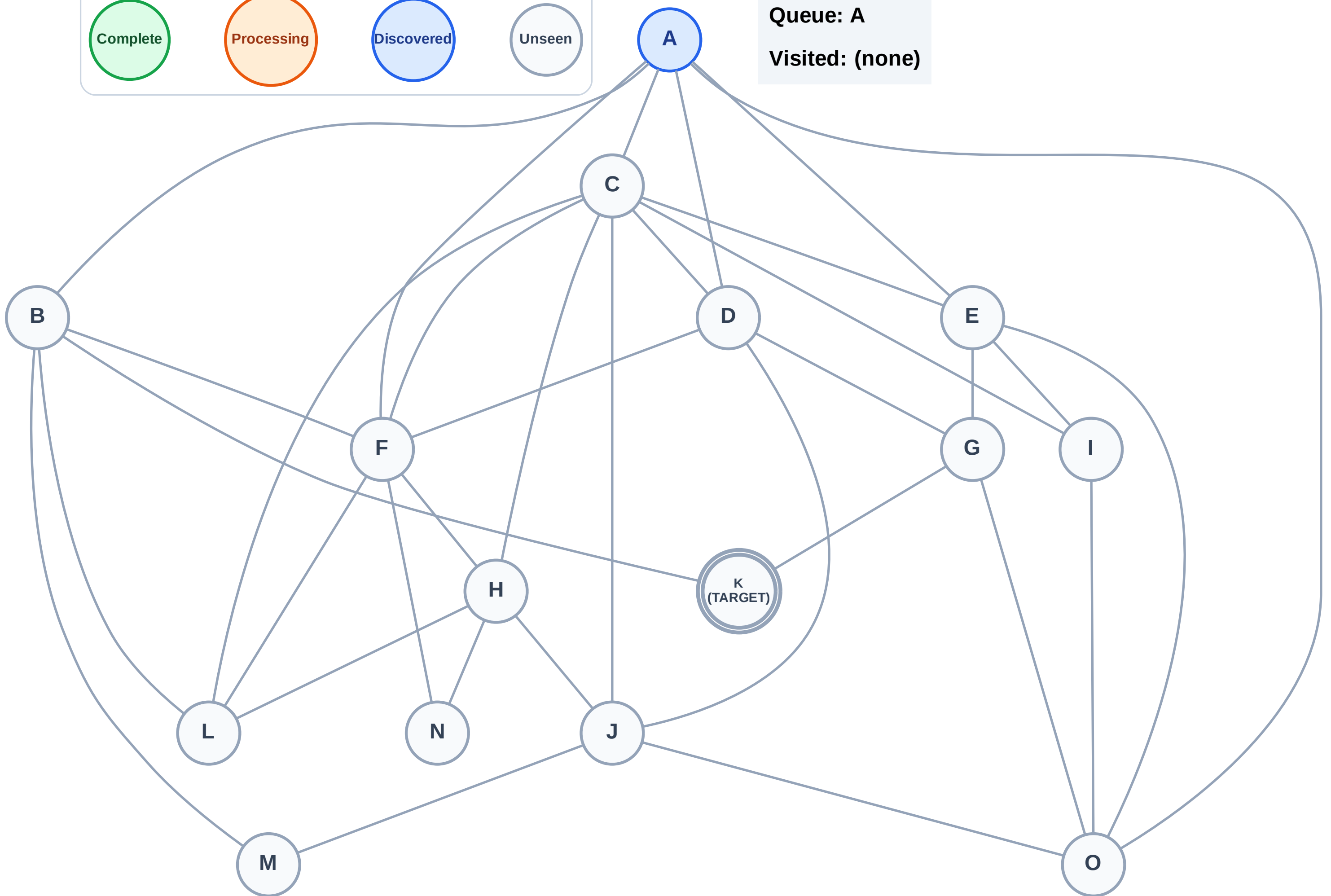
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

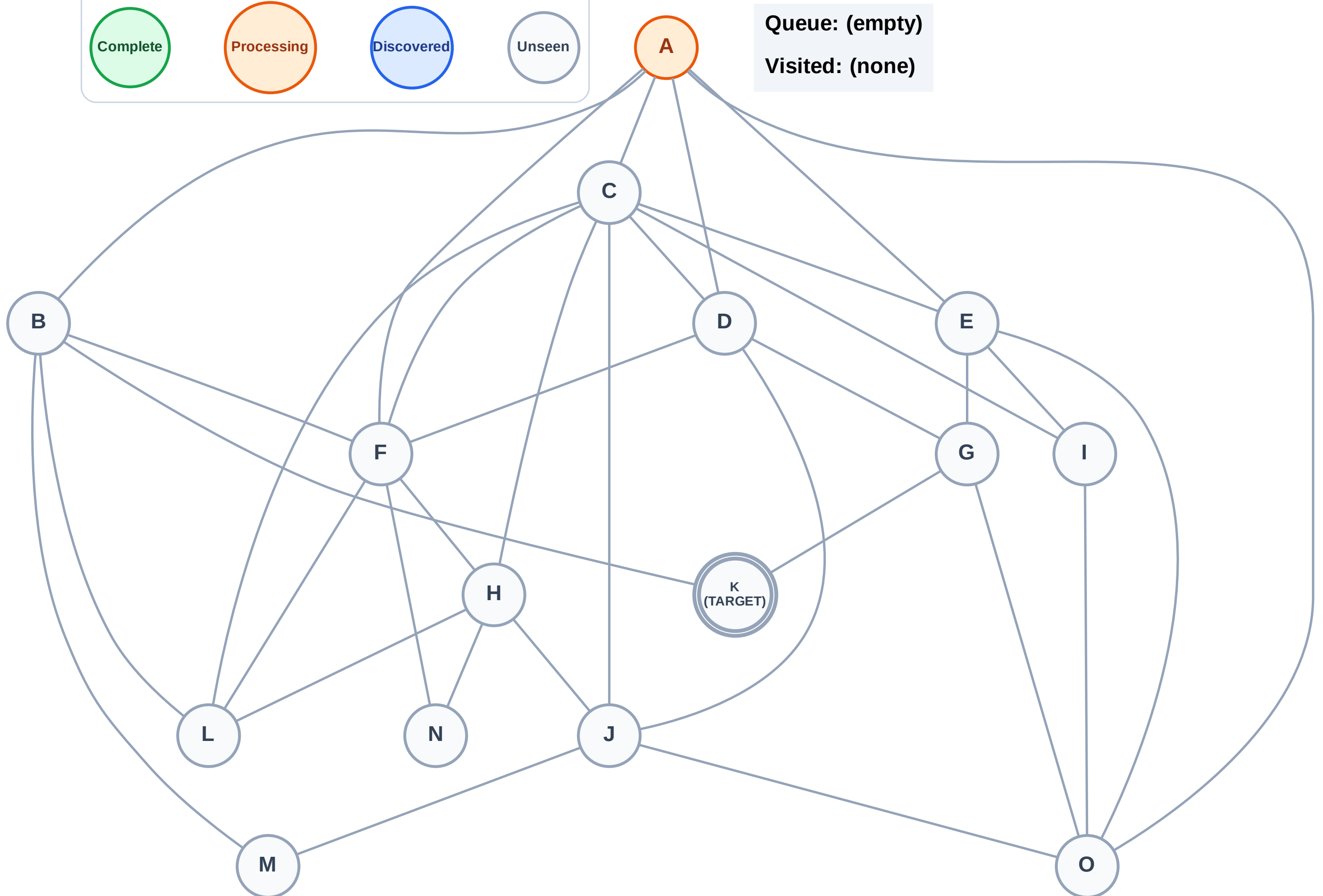
Step 2: Process node A

Legend



Queue: (empty)

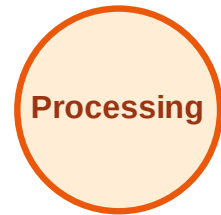
Visited: (none)



BFS Traversal

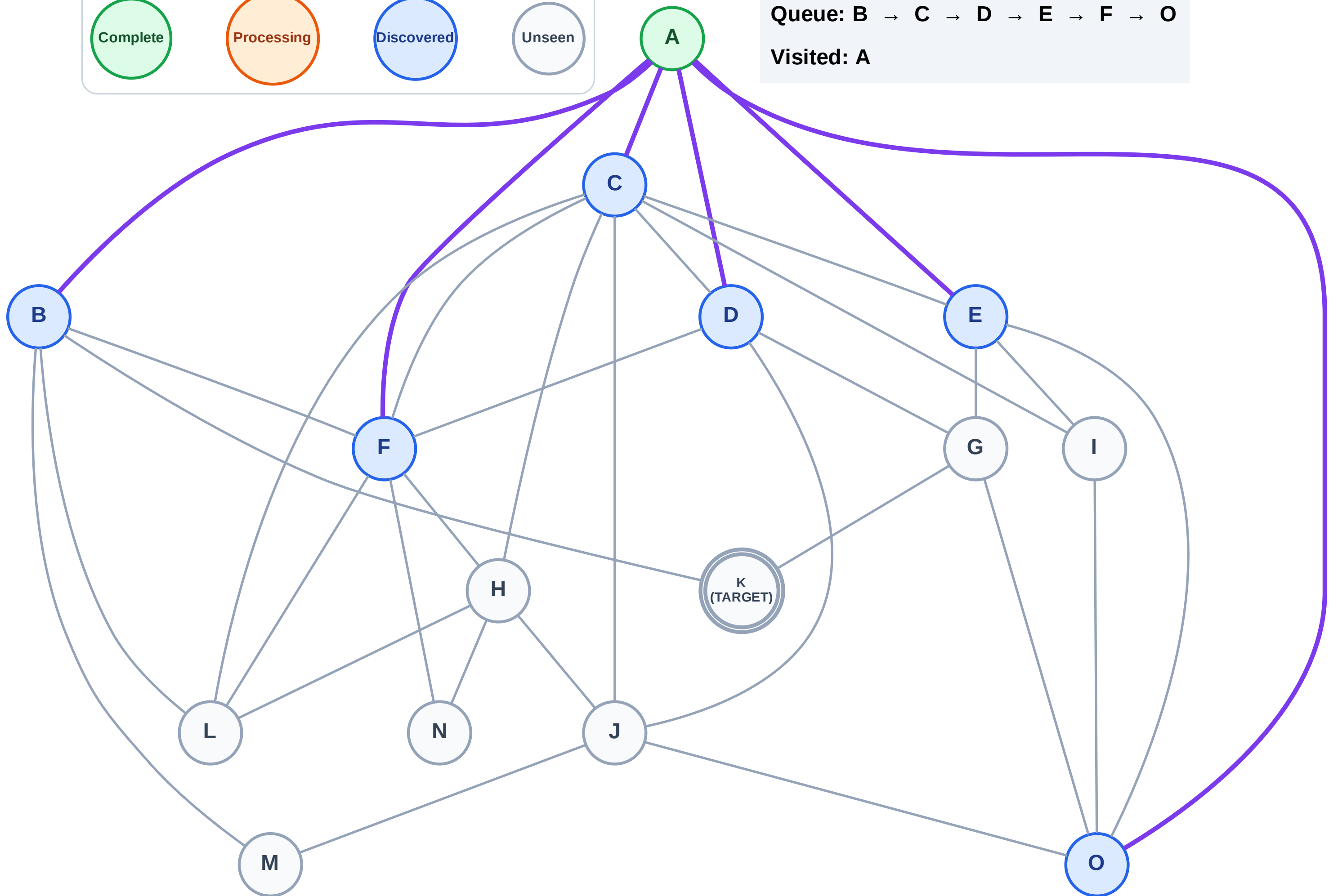
Step 3: Discover B, C, D, E, F, O from A

Legend



Queue: B → C → D → E → F → O

Visited: A



BFS Traversal

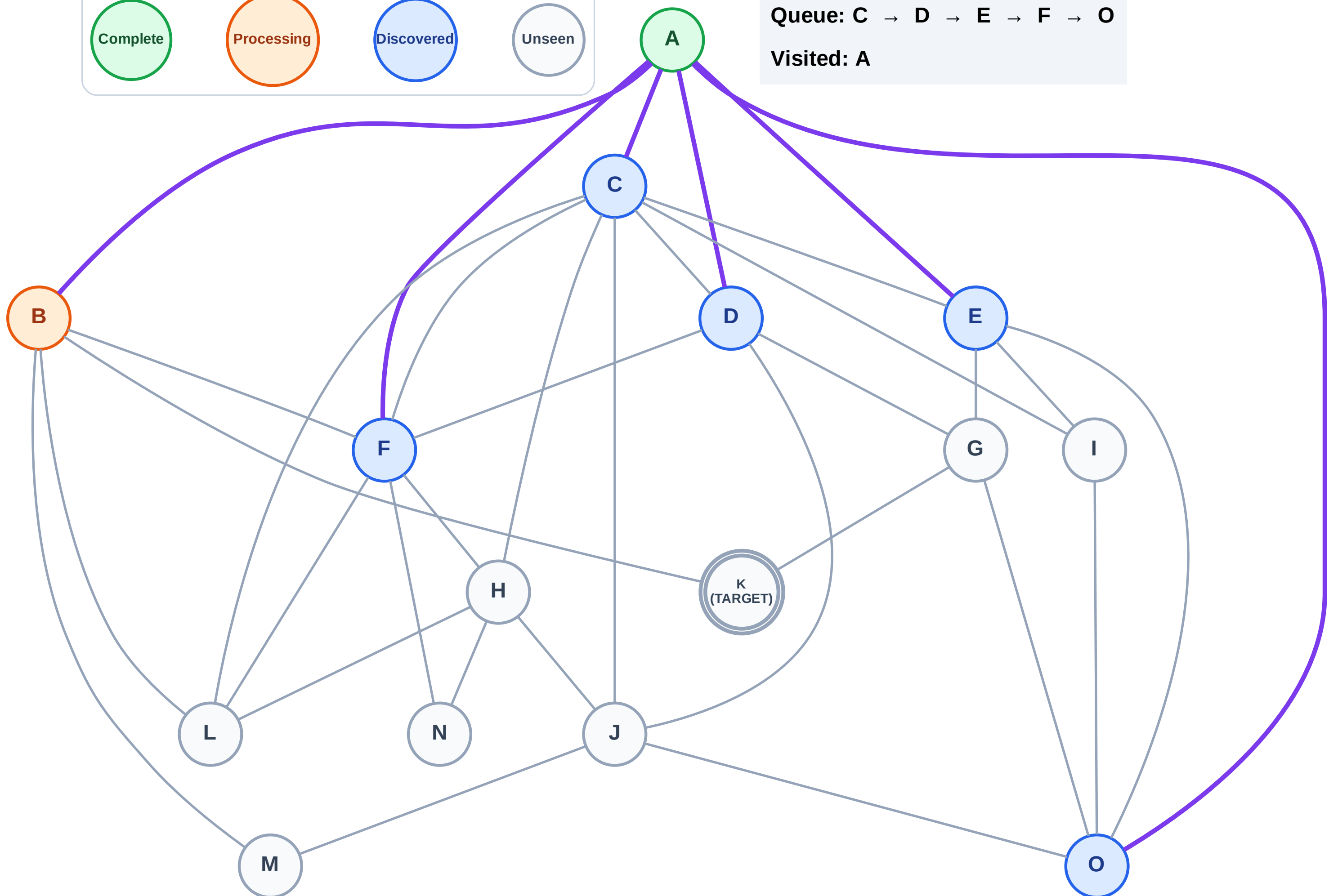
Step 4: Process node B

Legend



Queue: C → D → E → F → O

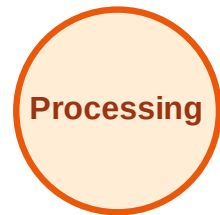
Visited: A



BFS Traversal

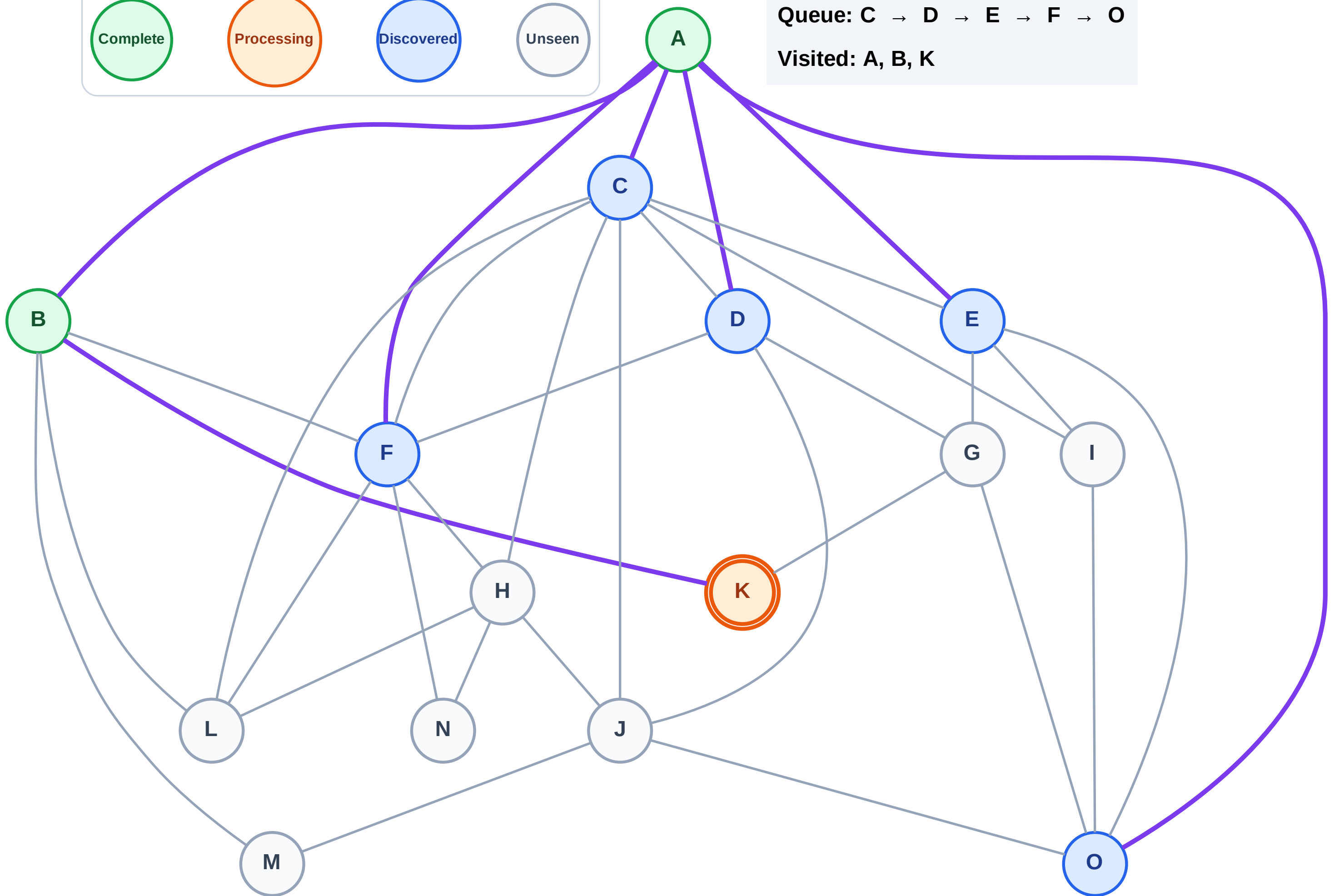
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: C → D → E → F → O

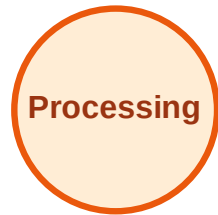
Visited: A, B, K



DFS Traversal

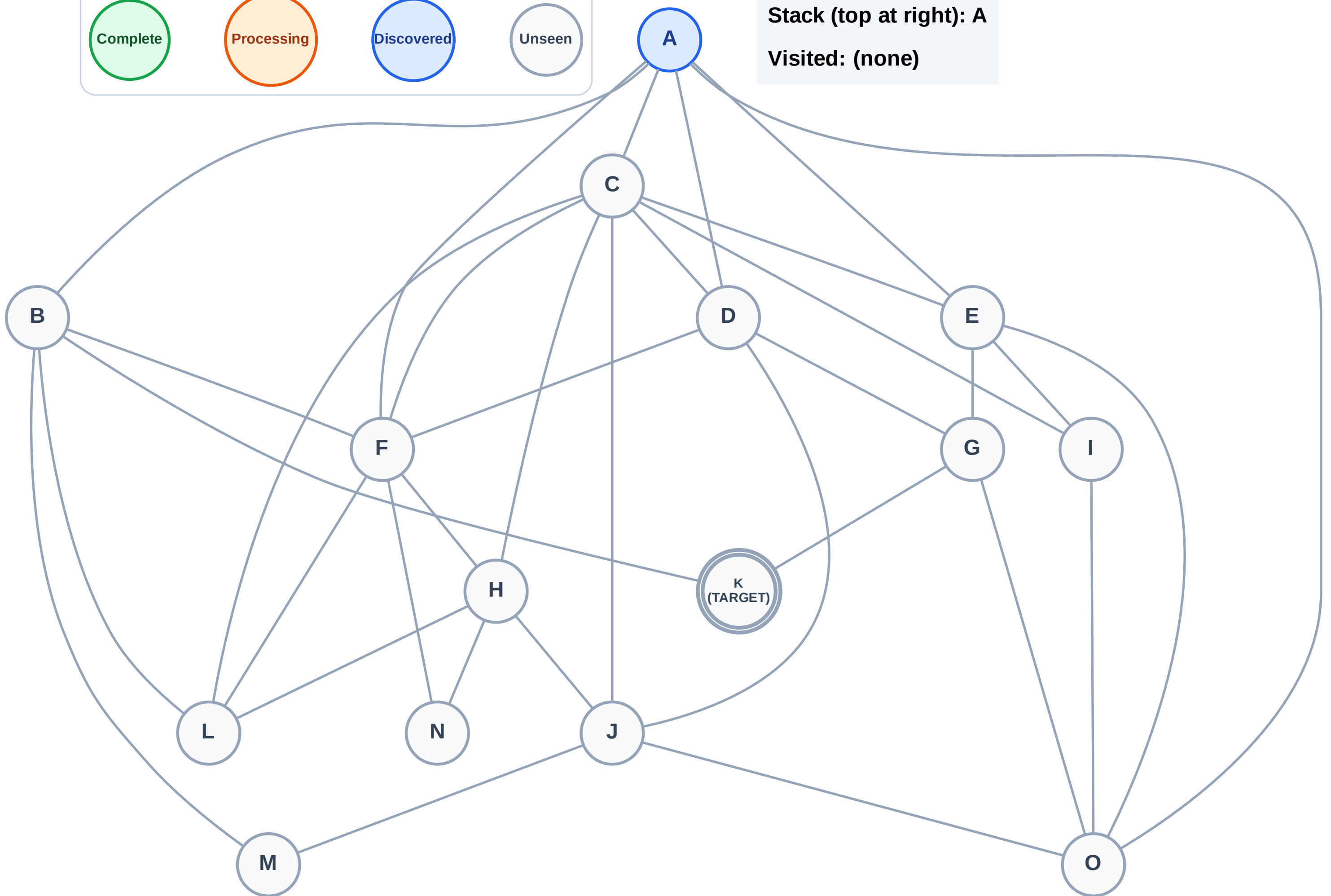
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

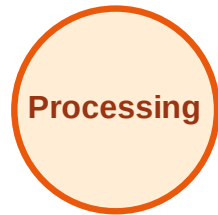
Visited: (none)



DFS Traversal

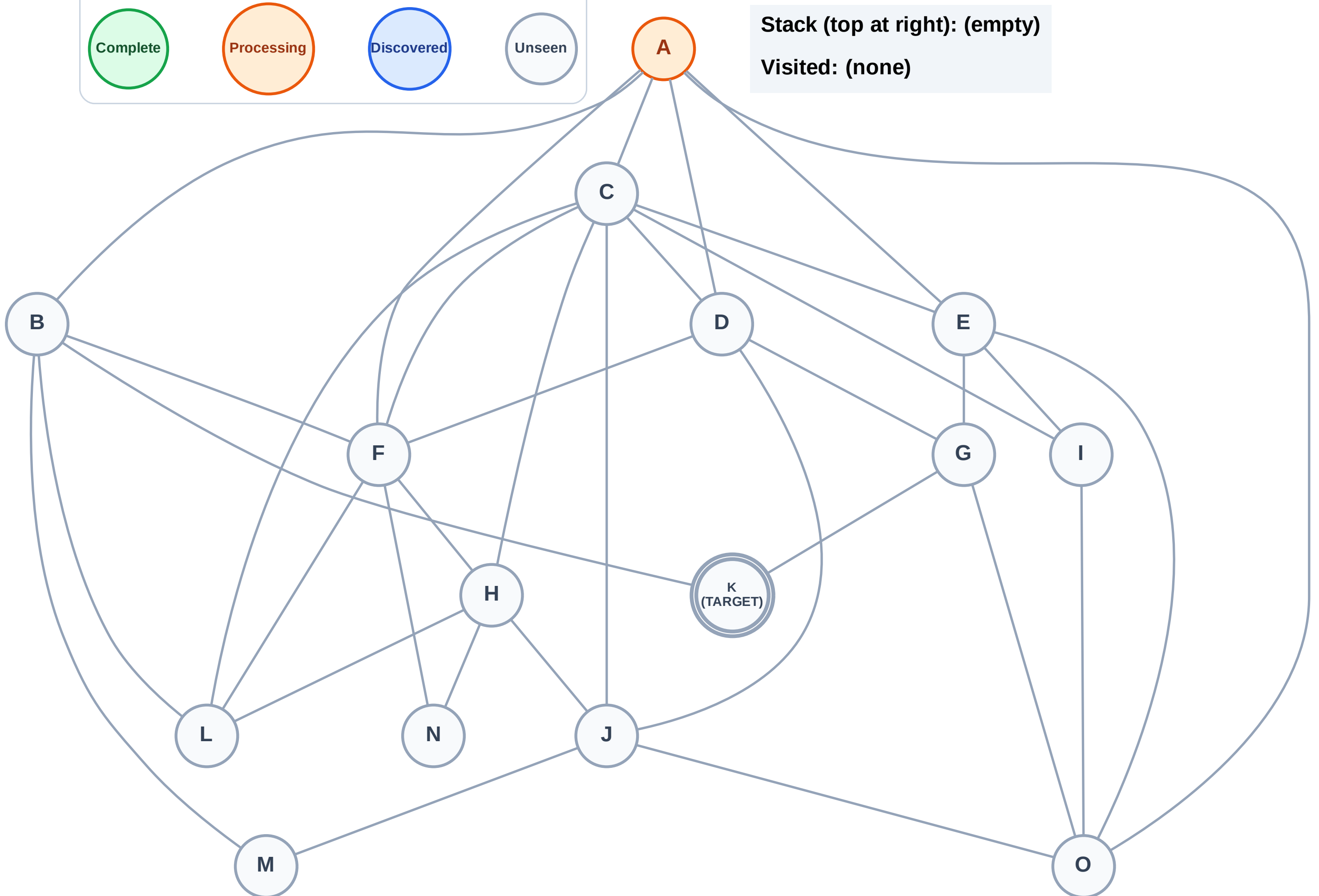
Step 2: Process node A

Legend



Stack (top at right): (empty)

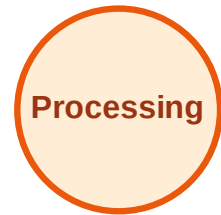
Visited: (none)



DFS Traversal

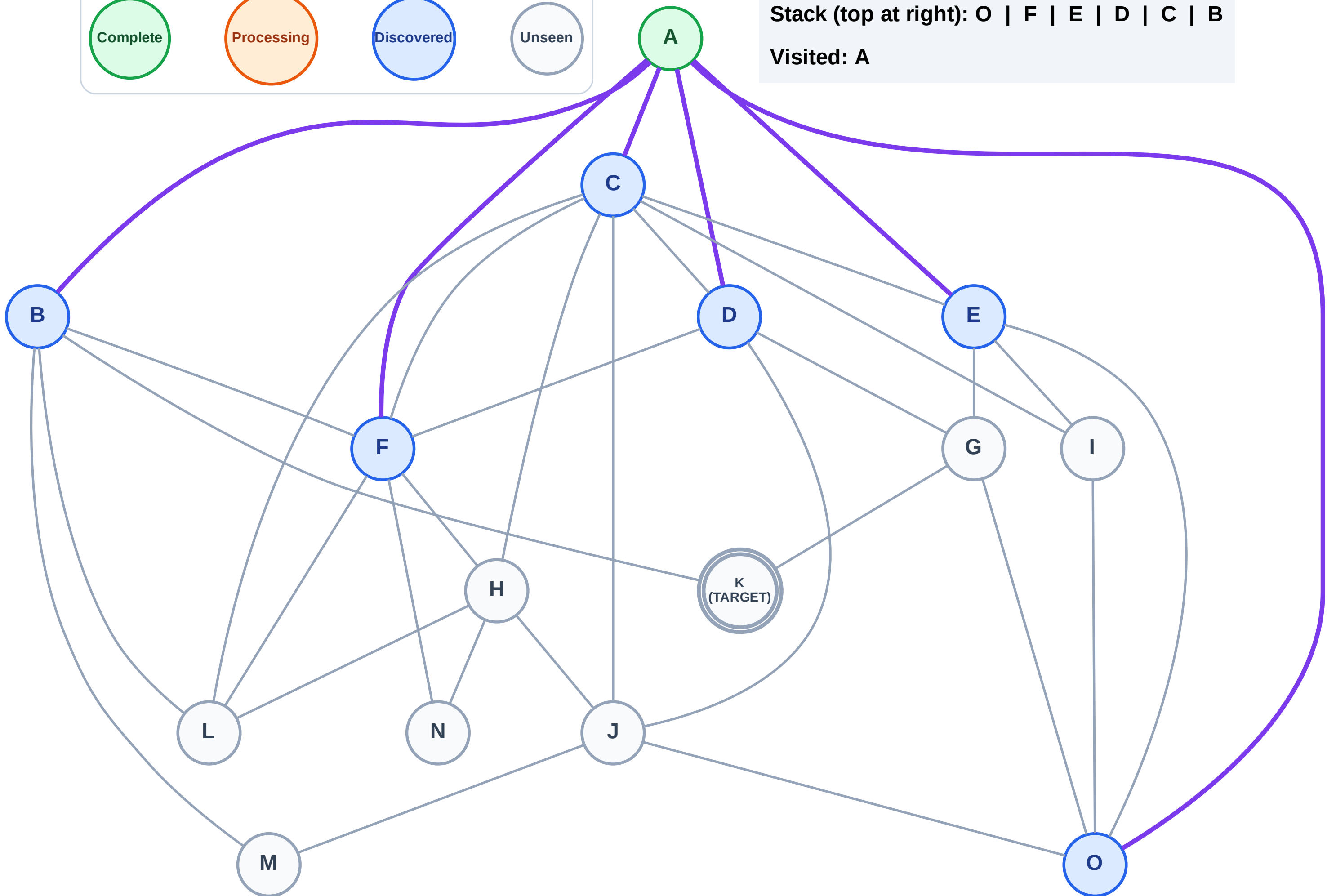
Step 3: Discover O, F, E, D, C, B from A

Legend



Stack (top at right): O | F | E | D | C | B

Visited: A



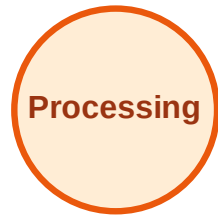
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



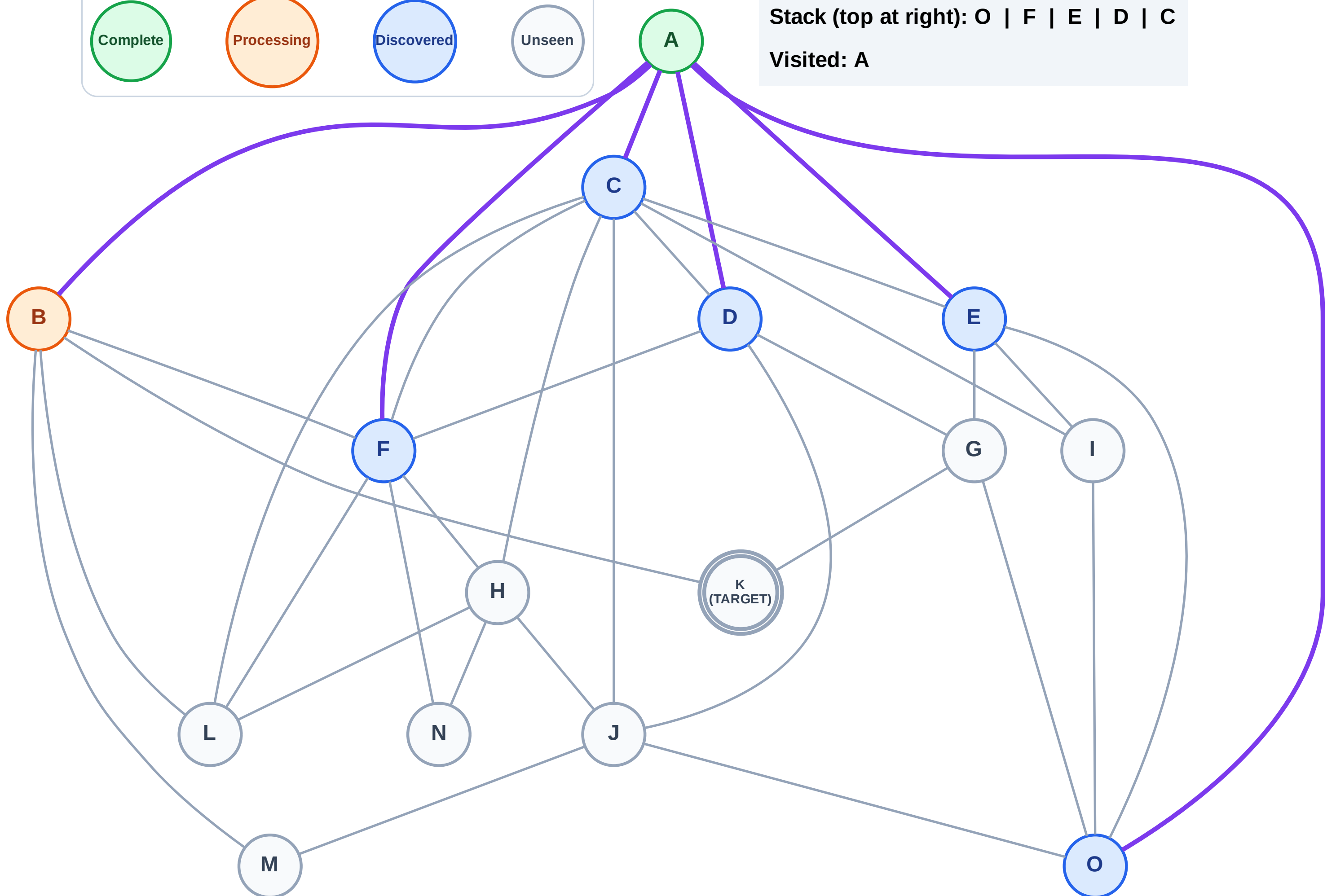
Discovered



Unseen

Stack (top at right): O | F | E | D | C

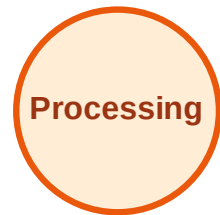
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): O | F | E | D | C | M | L

Visited: A, B, K

